GLANCED-IO: Taming I/O Optimization for Deep Learning at Scale

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Abstract

Scientific deep learning (DL) at scale typically trains on terabyte-scale datasets across thousands of accelerators, placing immense pressure on storage systems to keep pace with computation. Existing solutions respond to this demand by tuning individual I/O parameters to accelerate training performance. However, these techniques are limited by costly experiments, configuration space explosion, and inability to generalize application-specific optimizations. This leads to applications running with suboptimal configurations that reduce training efficiency, system utilization, or both. To address the challenge of finding the optimal configuration efficiently, we developed GLANCED-IO, a cross-layer I/O optimization framework that optimizes DL pipelines with high-fidelity approximation and efficient configuration space exploration. Through this work, we identified the following three key findings. First, independently optimizing either the application or system configurations leaves up to 2.4× performance on the table for scientists to efficiently run DL pipelines on HPC systems. Second, GLANCED-IO's one-factor-at-a-time (OFAT)-guided greedy exploration strategy achieved results comparable to more-expensive autotuning techniques while removing the pre-training required by ML-based approaches. Third, GLANCED-IO avoids executing the full application during optimization by operating on representative data subsets without GPUs, yet preserves 93% performance fidelity on average when deployed in DL pipelines. We demonstrate the efficacy of GLANCED-IO by optimizing large-scale global weather forecasting DL workloads, achieving up to 1.57× better performance than state-of-the-art with 2.3× fewer configuration evaluations than AIIO and 3.3× faster optimization than DeepHyper.

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CCS Concepts

• **General and reference** → **Performance**; • **Computing methodologies** → **Distributed artificial intelligence**.

Keywords

GLANCED-IO, Deep Learning, I/O Optimization, Autotuning, Joint Metric, Proxy Approximation, OFAT, DFTracer, DLIO

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1 Introduction

Training large-scale deep learning (DL) models for scientific applications such as genomics [53, 61], protein folding [39, 56], and

global weather forecasting [26, 27, 38, 46, 47] relies on terabyte-scale datasets that can reach petabyte-scale I/O over the course of training due to repeatedly loading the data across iterations. These applications typically run on hundreds to thousands of accelerators on large-scale HPC systems to keep training time tractable [38, 47]. The resulting data movement between storage systems and accelerators puts immense pressure on the I/O subsystem: training pipelines must continuously stream input samples from filesystems and periodically write checkpoints to preserve training state [47]. To manage this I/O complexity, modern DL frameworks provide highly configurable pipelines with tunable parameters across multiple software layers, such as data loader threading parallelism, prefetching depth, compression, storage backend selection, and many others [3, 43, 49]. Appropriate configuration of these parameters can yield substantial reductions in training time [34, 57], but the high dimensionality of this configuration space makes manual tuning impractical at scale [25, 36, 50].

Existing approaches address I/O performance primarily within two layers of the software stack: middleware and DL framework. At the middleware layers, systems accelerate individual storage operations through techniques such as caching (e.g., SHADE [41]), prefetching (e.g., noPFS [37]), and I/O optimizations (e.g., HVAC [40] and DYAD [32]), achieving up to 10× speedup of the training pipeline. These optimizations are transparent to applications and operate independently of application-level configuration.

Similarly, DL framework layers expose tunable parameters that can significantly impact end-to-end training time [15, 18] and are configured independently of the underlying storage system. Analogous to hyperparameter optimization in machine learning (ML), these parameters can be optimized using an exhaustive search. However, this approach is impractical for users with limited budgets [50], despite its ability to yield optimal results. As a result, practitioners adopt methods to reduce parameter space exploration time, including autotuning methods (e.g., DeepHyper [21, 36] and H5Tuner [24, 25]) that rely on repeated iterations, and ML-based methods (e.g., AIIO [35] and TunIO [50]). However, these approaches still require many costly workload execution samples or extensive training datasets, and may not be generalizable to unseen workloads.

Although existing approaches have shown success in improving performance in individual layers, they present three key limitations. First, existing techniques *tune knobs in isolation*, with some focusing solely on application performance [32, 37, 40] and others targeting system-level metrics [31, 33, 55], leaving up to 2.4× performance unrealized (as shown in Section 3) by not exploring both layers together. Second, jointly optimizing across layers leads to a *configuration space explosion*; navigating this space is prohibitively expensive, requiring either strict evaluation budgets on autotuners [21, 36] or massive datasets—such as the petabyte-scale data collected over years—for ML-based methods such as AIIO [35]. Both approaches also incur substantial runtime overhead for exploration or significant offline training costs, making them impractical for general usage for real-world DL workloads. Third, existing techniques usually require *running the full DL pipeline*, which might take hours to finish an iteration. This adds significant time and resource overheads across several hundreds of iterations, making DL pipeline optimization infeasible for scientists. Together, these three limitations necessitate a cross-layer optimization framework that strategically reduces the vast parameter space while forgoing full pipeline executions during optimization.

To address these challenges, we developed GLANCED-IO, a novel framework that optimizes DL I/O pipelines efficiently on large-scale HPC systems. GLANCED-IO consists of four components: *Performance Evaluator*, *Proxy Generator*, *Subset Selector*, and *Guided Optimizer* that allows users to optimize both the application and the system configurations to their fullest with minimal time and resource overheads.

To address isolated tuning, *Performance Evaluator* utilizes a multi-objective metric that accounts for both application throughput and system utilization instead of single-objective optimization, ensuring that performance never degrades even when headroom is limited. It also behaves as a validator for each translation point between application, proxy, and subset, guaranteeing that optimizations transfer back to the original workload.

To address configuration space explosion, *Guided Optimizer* uses one-factor-at-a-time (OFAT) exploration combined with a greedy algorithm by prioritizing high-impact parameters based on empirical findings from literature and documentation, reducing configuration space from exponential to linear. *Guided Optimizer* reduces the runtime overhead incurred by autotuners and avoids the pre-training cost of ML-based approaches, enabling efficient optimization of DL workloads and achieving similar or better optimization results than state-of-the-art techniques.

Finally, GLANCED-IO employs both *Proxy Generator* and *Subset Selector* to avoid running the full pipeline during optimization. *Proxy Generator* approximates full-pipeline behavior using GPU-free proxies built on DLIO [34], simplifying generation while preserving essential I/O characteristics and reducing resource consumption (GPU allocation, power) during tuning. *Subset Selector* further applies data-centric subsetting on top of proxy to run optimization at a fraction of full cost while preserving pipeline characteristics. Beyond reducing runtime cost, the combination of *Guided Optimizer* and *Subset Selector* also minimizes dataset versions and sample counts, reducing storage footprint during optimization. Evaluation on five weather forecasting workloads demonstrates that GLANCED-IO improves application throughput and I/O bandwidth by up to 1.8×, reduces configuration space by up to 2.3×, all without requiring prior data collection.

In summary, this work makes the following contributions:

(1) We develop a *multi-objective metric evaluator* that unifies application performance and system utilization, enabling holistic cross-layer I/O optimization.
(2) We design a *guided Greedy-OFAT optimizer* that tunes I/O parameters in linear time, matching or outperforming autotuning and ML-based approaches without requiring prior data collection.
(3) We build a *proxy-based approximator* that combines GPU-free proxies with data-centric subsetting to reduce optimization runtime to a fraction of full cost while maintaining 93% accuracy.

2 Background

2.1 Deep Learning Software Stack

Training DL models involves multiple software layers that move data from storage to accelerators. Frameworks such as PyTorch [49]

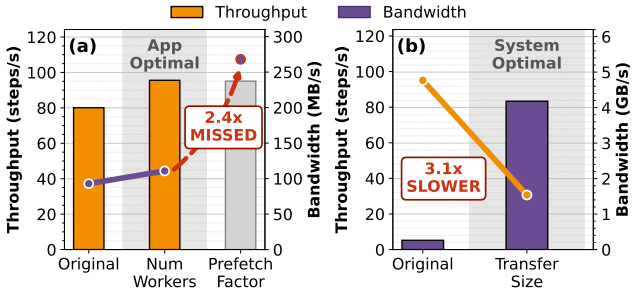


Figure 1: Impact of siloed metric optimization on `SFNO-SM`. (a) Optimizing application throughput alone stops at *num workers*, missing 2.4× system I/O bandwidth improvement in *prefetch factor* tuning. (b) Optimizing system I/O bandwidth by tuning *transfer size* slows throughput by 3.1×.

and TensorFlow [43] orchestrate computation and provide abstractions for model definition. Beneath the framework, the data loader layer manages parallel workers that fetch and preprocess samples, staging batches for GPU consumption. These workers use high-level I/O libraries (e.g., h5py [8] and NumPy [13]) to translate dataset operations into calls to low-level system libraries (e.g., POSIX and MPI-IO) that interact with the file system. This layered architecture also introduces heterogeneous resource dependencies: data loading alternates between CPU-bound preprocessing and I/O-bound storage access, training is GPU-bound, and distributed training adds communication-bound phases during gradient synchronization.

2.2 Performance Profiling

Optimizing DL pipelines requires visibility across system and application layers. System-level tools such as Darshan [55] capture I/O counters but lack application semantics, while application profilers such as PyTorch Profiler [17] and TensorFlow Profiler [12] provide timing for data loading and computation but do not trace underlying I/O operations. Cross-layer tracers such as DFTracer [33] address this gap by capturing both application events (e.g., data loading, preprocessing, computation) and I/O operations (i.e., read, write) with unified timestamps. From DFTracer traces, users can derive metrics such as application throughput and I/O bandwidth that serve as optimization objectives.

2.3 Configuration Parameters

The DL stack comprises at least three layers that expose tunable parameters affecting I/O performance: data loading, dataset, and storage. At data loading layer, practitioners configure the number of parallel workers and prefetch depth to control how aggressively batches are staged ahead of computation. At the dataset layer, choices include file format (e.g., HDF5 [7], TFRecord [16]), compression, transfer size and data layout. At the storage layer, parallel file systems expose striping configuration (e.g., stripe count, stripe size) that determine how I/O operations map to storage hardware. These parameters exhibit complex cross-layer dependencies. For instance, increasing worker count raises I/O concurrency but may cause CPU contention, while misaligned transfer sizes relative to stripe configuration can fragment requests and reduce bandwidth.

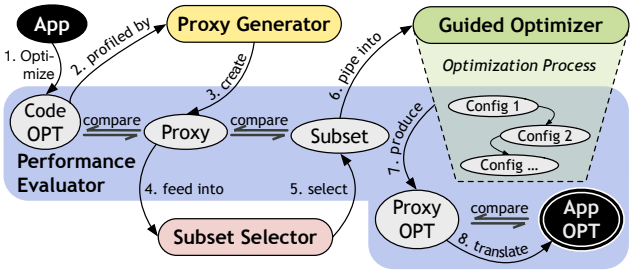


Figure 2: Design of GLANCED-IO.GLANCED-IO's design enables efficient and tractable end-to-end optimization for deep learning workloads.

3 Motivation

GLANCED-IO is designed to address three key challenges in state-of-the-art I/O tuning for DL workloads. First, existing approaches use *siloed metrics*, optimizing either application throughput or I/O bandwidth but not both. To illustrate, we run `SFNO-SM`, a global weather forecasting model (**Figure 1**). As shown in **Figure 1a**, optimizing for application throughput stops at the number of workers since further tuning yields diminishing returns, missing a 2.4× improvement in system I/O bandwidth. Conversely, **Figure 1b** shows that optimizing for system I/O bandwidth by tuning transfer size yields 15× improvement but degrades application throughput by 3.1×. This reveals a fundamental mismatch: I/O sizes that maximize system performance can be suboptimal for the GPU pipeline. Second, DL I/O pipelines suffer from *configuration space explosion*: $O(N^K)$ for K knobs with N values each. While smarter autotuners [20, 25, 36] navigate this space more efficiently, they still require evaluation budgets [21, 36] (e.g., configuration limits or time restrictions) that may prevent finding high-quality solutions. ML-based approaches such as AIIO [35] avoid exhaustive search but require expensive offline data collection and may fail to generalize across workloads. These approaches raise additional concerns: (1) when should data collection stop, given that more data may improve accuracy but increase collection cost; (2) what preprocessing is required, as these methods shift complexity from configuration search to model engineering (e.g., feature selection and model tuning); (3) how well can models generalize, given that workloads evolve and performance drifts, requiring continual maintenance [52]. These approaches also overlook storage footprint: methods such as DeepHyper [21] require the configuration space to be defined upfront. For I/O tuning, this means either pregenerating all dataset variants, which grows prohibitively for large-scale datasets, or generating on-the-fly, which inflates optimization time (**§4.5**). Third, current tools require *full DL pipeline execution* to evaluate each candidate configuration, incurring significant runtime and resource costs per evaluation. This highlights the critical need for data reduction and lightweight GPU-free proxies for optimization.

4 GLANCED-IO

This section outlines the design of GLANCED-IO. GLANCED-IO is a combination of orchestration scripts and analysis tools that addresses the limitations mentioned in Section 3 through four components: *Performance Evaluator*, *Proxy Generator*, *Subset Selector*, and *Guided Optimizer*, as illustrated in **Figure 2**. The steps are as follows: Application code will be ① optimized first, then *Proxy Generator* ② automatically profiles the code-optimized application

and ③ creates a proxy. Next, the proxy is ④ fed into *Subset Selector* which ⑤ automatically selects a representative subset. This subset then ⑥ pipes into the *Guided Optimizer* that ⑦ produces optimized configurations. Finally, the optimized configuration ⑧ is translated back to the original application. Each step is validated by *Performance Evaluator* to ensure all translation points are high-fidelity and improvements can be transferred back to original pipeline.

4.1 Code Optimization

In complex codebases such as DL pipelines, inefficiencies can come from the code itself. Despite having highly optimized DL framework, users can often write inefficient code during development that inherently affect the performance of the application, especially in the data loading layer. For instance, users often prototype with additional data, such as saving and loading more than necessary, even though this is not required in production. Another example arises from the flexibility of the framework itself, e.g., PyTorch encourages users to inherit the `Dataset` class and implement custom logic for data loading and preprocessing. Inefficiencies introduced at this layer can be easily overlooked and may propagate through the data pipeline. In particular, transferring unnecessary data can place pressure on shared memory, which serves as a synchronization point for parallel workers communicating with the main process.

To improve usability, DL frameworks also adopt generalized default behaviors that may not be optimal for all workloads. One such case is PyTorch's default collation step, which aggregates individual samples into a batch. When a workload already treats *each sample as a complete batch*, disabling collation avoids redundant processing and prevents unnecessary overhead. Another common issue in data pipelines is over-abstraction. Users often build on high-level frameworks or libraries layered on top of DL systems. While these tools offer intuitive APIs that accelerate prototyping, their internal complexity is substantial and their opinionated optimization strategies may not generalize across workloads [1, 2]. Since there is no single solution for every workload, GLANCED-IO encourages users to always manually optimize their codebase first to ensure inefficiencies do not inhibit the improvement of optimization stage.

4.2 Performance Evaluator

Performance Evaluator addresses the limitation of single-objective optimization by defining a unified multi-objective metric that accounts for both application throughput and system utilization. As all other components depend on performance metrics, *Performance Evaluator* serves as the foundation of GLANCED-IO. To enable this, GLANCED-IO analyzes code-optimized application traces and derives two metrics: application throughput (steps per second) and I/O bandwidth (bytes per second). We use application throughput rather than application time because throughput is inversely proportional to time while maintaining the same higher-is-better directionality as I/O bandwidth. These metrics validate fidelity across translation points and guide decision making during optimization.

Fidelity Validation. Because GLANCED-IO optimizes proxies and subsets, it must validate that performance characteristics are well-transferred at these three translation points: (1) proxy versus application, ensuring the proxy faithfully represents original I/O behavior, (2) subset versus full proxy, ensuring the subset preserves pipeline characteristics, and (3) optimized proxy versus optimized application, ensuring discovered configurations improve the original workload without degradation. We validate the quality of these translation points using the *fidelity* metric, defined as follows:

$$\text{accuracy} = \left(1 - \frac{|\text{target} - \text{baseline}|}{\text{baseline}}\right) \times 100\% \tag{1}$$

$$\text{fidelity} = 0.5 \times \text{accuracy}_{\text{throughput}} + 0.5 \times \text{accuracy}_{\text{bandwidth}} \tag{2}$$

where baseline refers to the reference point (e.g., full proxy) and target refers to the approximation (e.g., subset). Equal weighting ensures that neither metric dominates fidelity assessment, reflecting the cross-layer nature of GLANCED-IO. If fidelity falls below user-defined thresholds (e.g., this work uses 90% based on MLPerf Storage [4] recommendation), GLANCED-IO flags the configuration and prompts users to regenerate the proxy, adjust the subset, or proceed with caution.

Optimization Objective. We define multi-objective function by combining application throughput and system I/O bandwidth:

$$\text{score} = f(\text{throughput}, \text{bandwidth}) \tag{3}$$

Higher values in both metrics indicate better performance, allowing a simpler optimization objective. Furthermore, GLANCED-IO allows users to define their own objective functions (f). For example, we define f in this work as *weighted scoring*:

$$f = C \times \hat{t} + (1 - C) \times \hat{b} \tag{4}$$

where $\hat{t}$ and $\hat{b}$ are min-max normalized application throughput and I/O bandwidth, respectively, and C is a tunable priority constant.

4.3 Proxy Generator

Proxy Generator addresses the challenge of running full pipelines during optimization by creating lightweight and effective proxy applications. An effective proxy satisfies three criteria: (1) it must reproduce I/O access patterns of the original application, (2) it must preserve computational overlap between data loading and training, and (3) it must not require GPU resources. The first two criteria ensure that I/O optimizations discovered on the proxy transfer back to original application, while the third enables rapid evaluation at reduced cost. At its core, GLANCED-IO uses DLIO [34] as GPU-free proxy, albeit further configuration is necessary to satisfy above criteria. During this stage, given an application, *Proxy Generator* analyzes the traces to extract key I/O characteristics and instantiates DLIO configuration that mirrors these characteristics. *Proxy Generator* extracts the following characteristics from application:

Dataset Structure. The generator analyzes the original dataset to determine sample count, sample dimensions, file format (e.g., NPZ, HDF5, JPEG), and specific format configuration such as chunk size and compression in HDF5. It then creates a synthetic dataset within DLIO that matches these properties, ensuring that file sizes, access granularity, and metadata operations mirror the original.

I/O Access Patterns. DL workloads exhibit distinct access patterns depending on data loading strategy. Generator identifies specific I/O patterns such as single-file or file-per-sample layouts. It also captures prefetch depth and number of data loading workers from the profile. DLIO is then configured to reproduce these patterns to ensure matching with original workload.

Preprocessing Behavior. DL pipelines often include preprocessing stages such as normalization and augmentation that consume significant CPU resources and affect pipeline throughput. *Proxy Generator* analyzes per-sample preprocessing time from the traces and fits the statistical distributions to capture average behavior and variability. Then, *Proxy Generator* will configure DLIO to inject equivalent delays during data loading, preserving the balance between I/O and CPU in the pipeline. This modeling is critical because preprocessing often dominates data loading time, and ignoring it would yield proxies that overestimate I/O bottlenecks.

Computation Time. To capture computational overlap, the proxy must simulate the computation time spent by the application on forward pass, backward pass, and optimizer steps. To simplify the process, *Proxy Generator* only measures the overall per-batch computation time from traces and configures DLIO to inject synthetic compute delays of equivalent duration. These delays determine how aggressively the data loader can prefetch subsequent batches, directly affecting I/O concurrency and throughput.

Distributed Training Synchronization. Distributed DL training introduces synchronization points that affect overlap behavior between I/O and computation. These behaviors vary across GPU vendors and deployment, and are opaque to the application. Accordingly, *Proxy Generator* identifies synchronization behavior by analyzing the temporal relationship between data loading and batch boundaries. If analysis reveals data loading stalls during gradient synchronization, the generator configures DLIO to enable blocking synchronization (via `MPI.Barrier`) after each step, otherwise the generator models computation and communication as overlapped.

4.4 Subset Selector

Subset Selector addresses the cost burden of full evaluation runs by selecting a subset that still retains the pipeline behavior. However, there are several considerations to be made when selecting a subset. First, datasets from DL workloads are usually defined as logical mappings rather than explicit file lists (e.g., time-series data with temporal dependencies), complicating the identification of representative subsets—an irrelevant complication in I/O optimization. Second, different DL workloads might have different requirements, making one specific application approach not generalizable to other applications. *Subset Selector* simplifies the subsetting process by generating synthetic datasets using proxy, leveraging the convergence of application-system performance to a *steady state* before full-dataset traversal—enabling optimization on reduced data volume with proportionally lower execution time and storage footprint.

Strategy. Subsets closer in size to the original dataset tend to exhibit similar performance characteristics. As such, GLANCED-IO employs simple halving strategy to find the minimal representative subset. Starting from the full proxy dataset, the selector repeatedly halves the dataset size and validates fidelity at each step using *Performance Evaluator* (**§4.2**). The halving continues until either target size reduction is achieved or the fidelity drops below acceptable thresholds, whereupon the selector reverts to the previous size.

Benefit. Beyond reducing runtime cost, subsetting also minimizes storage footprint. During optimization, users may explore multiple dataset configurations (e.g., different layouts and compressions), each requiring a separate copy of dataset. Operating on subsets reduces storage requirements proportionally, enabling exploration of more configurations within fixed storage budgets.

Table 1: GLANCED-IO cross-layer optimization parameters.

	Layer	Optimization	
More Portable ↓	App	Num Workers	Easier ↓
		Prefetch Factor	
		Dataset Access Pattern	
Less Portable	System	PFS Striping	Harder
		Transfer Size	

Caveat. Although aggressive subsetting will reduce optimization time significantly, it can introduce unintended side effects that compromise fidelity. When subsets become small enough to fit in file system caches or memory, I/O bandwidth measurements spike artificially as reads are served from cache (e.g., DRAM) rather than underlying storage. This alters I/O behavior, which in turn affects the overlap between data loading and computation, causing deviation of both application throughput and I/O bandwidth measurements from the full dataset. To address this, *Subset Selector* uses *Performance Evaluator* to catch these issues through fidelity scoring, prompting users to revert to a larger subset size.

4.5 Guided Optimizer

Guided Optimizer addresses the exacerbated configuration space and optimization cost, due to joint assessment of application and system parameters, by reducing the space rather than simply navigating the space efficiently. For this, *Guided Optimizer*, a Greedy-based autotuner, internally uses high-impact parameters based on empirical findings and software documentation while also employing one-factor-at-a-time (OFAT) to reduce the configuration space.

Parameter Ordering. As shown in **Table 1**, GLANCED-IO ranks parameters from more portable and easier to tune (top) to less portable and harder to tune (bottom). First, *Guided Optimizer* starts with application-layer parameters, bootstrapping optimization with accessible tuning knobs. This design choice democratizes I/O optimization for users without storage expertise. For instance, number of workers and prefetch factor are built into DL frameworks such as PyTorch and TensorFlow, requiring no system expertise to modify. Similarly, dataset access pattern can easily be adjusted by modifying the dataset itself (e.g. enabling sequential reads or changing the format, compression and the layout of the data). Next, for users who can leverage system-level optimizations, such as parallel file system configuration that is widely used in HPC systems, GLANCED-IO then explores PFS striping followed by transfer size, which is aligned based on stripe size as recommended in DLIO [34]. Optimizing system-level factors can yield substantial improvements as they will align I/O operations with the underlying storage architecture. However, these parameters are platform-specific, system-specific, and dataset-specific, making them less portable across deployment environments. For example, PyTorch by default does not have the capability to modify dataset transfer size as users tend to choose their dataset format. Meanwhile, TensorFlow actively uses TFRecord [16] format that has `buffer_size` parameter, enabling the tuning of I/O transfer size during data loading. By

Algorithm 1: Guided Optimizer: OFAT with Greedy Selection

Input: Parameter $P = [p_1, \ldots, p_k]$, value pools $V = \{V_1, \ldots, V_k\}$, initial configuration C_0
Output: Optimized configuration C^*

1 $C^* \leftarrow C_0$;
2 best_score $\leftarrow$ PerformanceEvaluator(C^*);
3 **foreach** $p_i \in P$ **do**
    // OFAT exploration
4 improved $\leftarrow$ false;
5 **foreach** $v \in V_i$ **do**
        // Parallelize via job scheduler
6 $C' \leftarrow C^*$ with $p_i \leftarrow v$;
7 score $\leftarrow$ PerformanceEvaluator(C');
8 **if** *score* > *best_score* **then**
            // Greedy selection
9 $C^* \leftarrow C'$;
10 best_score $\leftarrow$ score;
11 improved $\leftarrow$ true;
12 **if** $\neg$*improved* **then**
            // Termination on plateau
13 **break**;
14 **return** C^*;

exploring portable parameters first, GLANCED-IO ensures that all users benefit from optimization even if system-specific tuning is inapplicable to their environment.

Algorithm. Given the ordered parameters, Algorithm 1 details the optimization workflow using OFAT. Rather than exploring all parameter combinations, OFAT varies one parameter while holding others fixed, identifies the best value for that varied parameter, and then moves on to the next parameter. This approach reduces the number of evaluations from $O(N^K)$ to $O(K \times N)$ for K parameters with N values each, making optimization tractable within reasonable time budgets. The algorithm uses streaming Greedy selection: as each value for one parameter is evaluated, *Guided Optimizer* immediately updates the best configuration for that parameter if improvement is observed (lines 7-10).

Storage Implication. Some stages in DL pipeline optimization (see **Table 1**), such as dataset access pattern, PFS striping, and transfer size, modify the data layout and consequently generate multiple copies of the dataset, quickly scaling storage to terabytes or even petabytes. At this scale, optimizing one parameter at a time is more tractable, since acquiring storage to accommodate multiple dataset copies is challenging even at large HPC sites [5, 6, 14]. Using OFAT, multiple copies of the dataset exist only within the explored parameter pool, rather than across the full combinatorial space of all parameters, thereby reducing the storage footprint.

Additional Optimizations. Several optimizations further reduce the cost of the *Guided Optimizer*. First, the algorithm can terminate early when a parameter yields no improvement, indicating diminishing returns (line 11), avoiding unnecessary evaluation. Second, because OFAT is deterministic and fixes one factor at a time, the algorithm can reuse observations across parameter explorations. For example, if the *Guided Optimizer* explored number of workers with values of 1-32 while prefetch factor was fixed at 2 and found 16 workers as the optimal configuration, the subsequent exploration of prefetch factor can skip evaluating prefetch factor 2 since that configuration was already assessed. Third, we acknowledge that OFAT does not capture all cross-parameter interactions: strongly coupled parameters may require joint tuning to reach the best configuration. In our workloads, however, these effects were not prominent enough to justify broader combinatorial search. If such interactions do arise, OFAT naturally supports lightweight refinement: after the initial sweep identifies promising regions, GLANCED-IO can selectively re-evaluate suspected coupled parameters while keeping the remaining parameters fixed. This preserves OFAT's scalability while recovering coordinated improvements at low additional cost.

4.6 Implementation Details

GLANCED-IO consists of orchestration scripts for workflow coordination, configuration files for proxy generation, and analysis utilities for processing profiling data, rather than a monolithic application. We describe the key implementation aspects below.

Profiling and Analysis. GLANCED-IO integrates with DFTracer [33] to trace both original applications and proxy runs, capturing per-operation timestamps, data sizes, and file access patterns in trace format. Furthermore, GLANCED-IO employs analysis involving large-scale traces from distributed runs, implemented on top of C++ and Python, utilizing scalable aggregation techniques similar to WisIO [58].

Framework Agnostic. GLANCED-IO operates on application and I/O behavior extracted from profiling traces rather than from inspection of application source code. This design makes GLANCED-IO conceptually compatible with any DL framework, including PyTorch, TensorFlow, and JAX. As long as the framework's computation and data loading pipeline can be annotated and the I/O calls can be traced with DFTracer, GLANCED-IO can generate proxies, select subsets, and optimize configurations without framework-specific modifications.

Proxy Generation. Proxy Generator produces YAML configuration files that instantiate DLIO with parameters matching the target application's data loading behavior. These configurations specify dataset structure, access patterns, preprocessing delays, and computation time derived from DFTracer traces. Users invoke DLIO with the generated configuration to run proxy evaluations.

Parallel Evaluation. With OFAT exploration strategy, evaluation of different values for the same parameter are independent. Hence, GLANCED-IO can dispatch them in parallel using job schedulers such as Flux [19] or SLURM [59]. Each configuration runs as a separate job, and results are collected upon completion for greedy decision-making before proceeding to the next parameter. This parallelization reduces wall-clock optimization time proportionally to the number of available compute nodes.

Platform Support. Although GLANCED-IO is conceptually platform agnostic, this work targets HPC environments with parallel file systems (e.g., Lustre [11, 29], GPFS [9]) that are widely used for large-scale DL training. As such, the current implementation interfaces with Lustre APIs to query and configure system-level parameters such as stripe count and stripe size.

Table 2: Workloads used in GLANCED-IO.

Workload	Nodes	Dataset Size	Duration (5 Epochs)
STORMER-LG	32	31 TB	~1h 24m
SFNO-LG	16	31 TB	~2h
TinySTORMER-LG	2	5.4 TB	~1h 26m
STORMER-SM	2	71 GB	~1h
SFNO-SM	1	71 GB	~1h

5 Evaluation

This section evaluates GLANCED-IO's effectiveness in optimizing I/O performance for distributed DL workloads. We address three questions aligned with our contributions:

(1) **Metric effectiveness.** Does a multi-objective metric that captures application performance and system utilization enable better optimization decisions than a single metric?
(2) **Proxy-based approximation.** Can proxy-based optimization reduce optimization cost while preserving sufficient fidelity for effective configuration selection?
(3) **Optimization effectiveness.** Can GLANCED-IO efficiently identify high-quality configurations and outperform baselines without requiring prior data collection?

We evaluate GLANCED-IO on five global weather forecasting workloads spanning different model architectures, dataset sizes, and node counts. The evaluation proceeds in five stages, aligned with these questions. First, we evaluate the performance improvement of optimizing pipeline code. Second, we demonstrate why joint application-system metric is necessary. Third, we assess the accuracy and cost reduction enabled by proxy-based approximation. Fourth, we evaluate optimization effectiveness across workloads and compare against baselines. Finally, we present `STORMER-LG` as a representative end-to-end case study. To avoid repetition, we summarize recurring patterns across remaining workloads compactly.

5.1 Experimental Setup

5.1.1 System Configuration. All experiments were conducted on Tuolumne at Lawrence Livermore National Laboratory (LLNL). Each node is equipped with 4 AMD MI300A GPUs, dual 4th-generation AMD EPYC CPUs (96 cores, 192 hyperthreads), and 512 GB of memory. Nodes are interconnected via HPE Slingshot 11 and shared Lustre parallel file system with a maximum bandwidth of 240 GB/s and a storage budget of 100TB.

5.1.2 Datasets. We use the ERA5 reanalysis dataset in HDF5 format [7] at two resolutions: 0.25° for ERA5-LG and 5.6° for ERA5-SM. Both datasets span 40 years (1979–2018), with each year partitioned into approximately 1,460 HDF5 files. Following the training configuration used by STORMER [47], each training step loads both input and target atmospheric variables from 69 HDF5 datasets per file. ERA5-LG comprises approximately 31 TB across 58,400 files, each containing 130 datasets with dimensions 721×1440 in `float32`, resulting in ~573 MB of data per step. ERA5-SM comprises ~71 GB across 58,300 files, each containing 114 datasets with dimensions 32×64 in `float32`, resulting in ~1 MB of data per step.

5.1.3 Workloads. We evaluate three global weather forecasting model architectures: SFNO [27], STORMER [47], and TinySTORMER, a reduced-parameter variant of STORMER. All models are implemented in PyTorch using Distributed Data Parallel (DDP) and use h5py to read HDF5 datasets. To train the models, we use the modified STORMER pipeline [45] such that SFNO can still use the same codebase to simplify the experiments. Throughout this section, each workload configuration is named `MODEL-DATASET`, where `MODEL` denotes the architecture and `DATASET` indicates the dataset size (`LG` for ERA5-LG, `SM` for ERA5-SM). Specifically, `TinySTORMER-LG` uses gradient accumulation and a subset of ERA5-LG due to memory and runtime constraints at smaller node counts. During proxy-based optimization, GPU execution is disabled and workloads are evaluated using CPU-only proxy runs. **Table 2** summarizes all evaluated configurations along with their 5-epoch durations, which are used to capture stable and representative I/O behavior. During experiments, all performance metrics are computed on a per-epoch basis. For each evaluated configuration, we run five epochs and report the average values. We observe less than 5% run-to-run variability and therefore we omit error bars throughout this evaluation.

5.1.4 Baselines. We compare GLANCED-IO with three baselines:

MaxResource reflects a common HPC practice in which users allocate and utilize the maximum available resources under the assumption that more resources yield better performance. For application configuration, we cap the number of workers based on the ratio of CPU hyperthreads to GPUs (48 workers per GPU on Tuolumne). We then search for worker and prefetch factor combinations that maximize memory usage, allocating up to 90% of total memory to workers to account for runtime overhead. For dataset configuration, MaxResource maximizes transfer size by packing datasets into a single HDF5 file. For system configuration, it uses all Lustre OSTs with stripe size matching the dataset transfer size.

AIIO [35] is an ML-based approach that predicts I/O bandwidth from low-level I/O counters. Since AIIO does not directly support DL pipelines parameters, we extend it using a two-stage inference process. The first stage predicts I/O counters from application configurations, and the second stage predicts system-level I/O bandwidth from the estimated counters. To train these models, we used traces across diverse workloads collected in DLIO [34] and WisIO [58]. Furthermore, since those collected data do not include our workloads defined in Section 5.1.3, we incorporate data from our optimization stages of GLANCED-IO to ensure the generality of the model. In summary, we used 7 days' worth of traces, 271 unique workloads, across 2,474 nodes, resulting in 1.2 billion events, to train AIIO model. After training, AIIO performs exhaustive search over the configuration space and selects the best configuration with the highest predicted I/O bandwidth.

DeepHyper [21] is an open-source Python package for scalable hyperparameter search. Although its main purpose is hyperparameter optimization, autotuning work on HEPnOS [36] shows that it can also be used to identify optimal storage software configurations. It uses a surrogate model to learn the relationship between configurations and the objective function, enabling efficient exploration of the search space. In this work, we used random forest regressor similar to HEPnOS autotuning and specify 12 hours budget for running the optimization stage. Similar to HEPnOS autotuning, we used application metric (throughput) as the objective metric.

Table 3: Code optimization achieves up to 1.24× improvements in application throughput and I/O bandwidth.

Application	Startup	Preprocess	Get Item	Thpt.	Bandw.
STORMER-LG	1.95×	1.46×	1.27×	1.11×	1.09×
SFNO-LG	1.82×	1.43×	1.14×	1.24×	1.23×
TinySTORMER-LG	1.66×	1.25×	0.92×	1.04×	1.04×
STORMER-SM	1.52×	1.01×	1.24×	1.11×	1.09×
SFNO-SM	0.81×	0.86×	1.03×	1.14×	1.13×

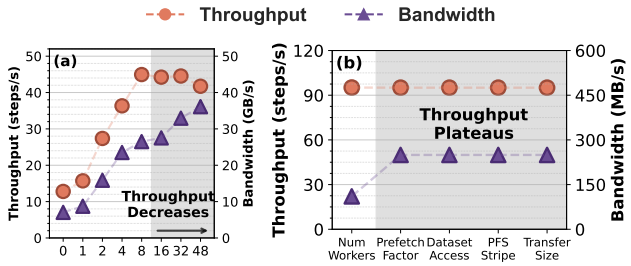


Figure 3: Single-metric optimization failures. ***(a)*** *Worker counts for* SFNO-LG *shows that optimizing only on system I/O bandwidth can obscure application throughput degradation.* ***(b)*** *Optimization on* STORMER-SM *shows that early termination on application metric can miss holistic pipeline gains.*

5.2 Impact of Code Optimization

To observe the impact of optimizing the original application's code, we run 5 epochs of all workloads for both original codebase and code-optimized codebase and compare their performance. We manually optimize trainer and data loading implementation of STORMER by (1) eliminating runtime overhead, such as unnecessary function calls and over-abstraction of the code, (2) reducing small I/O accesses, such as iterating directories to look up the dataset files, (3) minimizing unnecessary preprocessing, such as removing collate operation, and (4) eliminating unnecessary data movement, such as sending unnecessary variables to shared memory used for synchronization between parallel workers and main process.

As shown in **Table 3**, we can speed up the pipeline startup time by up to 1.95× simply through optimization of the codes. Notably, the effects of code optimizations are more pronounced in larger workloads as smaller workloads would obscure problems associated with data processing (i.e., preprocessing large dataset). Even though some of the internal application metrics such as pipeline startup, preprocess, and getitem time fluctuate across workloads, the impact towards application throughput and bandwidth still reaches up to 1.24×. This underscores the intricacy of DL pipelines, where optimizing a specific application layer may appear counterproductive, yet enables substantial gains at the pipeline level.

Takeaway 1. *Code optimization is necessary as performance bottlenecks can be hidden at small scales of data and deployments.*

5.3 Impact of Joint Metrics Optimization

To understand the importance of joint metrics, we run GLANCED-IO optimization stage across all workloads. Since GLANCED-IO optimizes application and system layers simultaneously, we monitor optimization progress and simulate early termination when one layer reaches a plateau, regardless of continued improvement in the other. **Figure 3** reveals that such early stopping can cause the optimizer to miss substantial gains in holistic pipeline performance. In the figures, x-axis shows the configuration currently being evaluated, left y-axis shows the application throughput and the right y-axis shows the I/O bandwidth. The progression of these metrics are shown as orange circle and purple triangle, respectively.

Optimizing system utilization alone can degrade application performance. Figure 3a shows worker-count sensitivity for SFNO-LG. As the number of data loader workers increases, I/O bandwidth improves monotonically, indicating higher filesystem utilization. However, application throughput peaks at a small worker count and degrades thereafter due to increased CPU contention. An optimization strategy that prioritizes system utilization alone would select configurations that reduce application throughput.

Optimizing application throughput alone can hide underutilized system resources. Figure 3b shows stage-wise optimization behavior for STORMER-SM on two nodes. After application-level tuning, application throughput plateaus, suggesting limited optimization headroom. However, system-level I/O bandwidth continues to improve across subsequent stages. An optimizer that relies only on application throughput would stop early and miss opportunities to improve system utilization without degrading execution.

Implications for cross-layer optimization. Across workloads and scales, application-level behavior and system-level utilization can diverge in both directions. Optimizing either perspective in isolation therefore leads to suboptimal outcomes. A joint metric is therefore required to reconcile these trade-offs and guide optimization toward configurations that improve end-to-end throughput while maintaining efficient system utilization.

Takeaway 2. *Single-metric optimization fails to account for cross-layer interactions in DL I/O pipelines, motivating a joint metric that aligns application and system performance.*

Table 4: GLANCED-IO achieves 93% average fidelity when generating proxies and subsets across diverse workloads.

Workload	App →Proxy	Full →Subset	Proxy →App
STORMER-LG	96.5%	97.7%	90.8%
SFNO-LG	94.8%	98.4%	54.6%
TinySTORMER-LG	79.5%	98.6%	94.1%
STORMER-SM	98.6%	99.6%	97.9%
SFNO-SM	99.0%	99.5%	96.1%

5.4 Fidelity of Proxy-Based Approximation

To understand the benefit of GPU-free proxy-based approximation during optimization, we annotated and traced five workloads performance using DFTracer and use *Proxy Generator* to build DLIO proxy and generate the synthetic data that mimic the I/O performance of the pipeline. Then, we keep configuration of the proxy the same as the original pipeline. Furthermore, these proxies will be processed further using *Subset Selector* to create subsets of synthetic datasets. To ensure translatability from proxy and subset, we examine three aspects: App-to-Proxy, subset, and Proxy-to-App.

App-to-Proxy Fidelity. Table 4 reports App-to-Proxy translation accuracy across all workloads. For large-scale workloads, accuracy exceeds 94%, and for small-scale workloads it exceeds 98%, indicating that the proxy closely matches application-level I/O

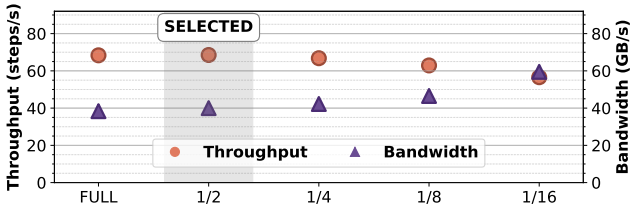


Figure 4: Effect of dataset subset size on I/O behavior for `STORMER-LG`. *Using half the dataset preserves full-dataset I/O behavior. Smaller subsets (1/8 and 1/16) enter a cache-dominated regime that inflates application throughput and alters I/O bandwidth trends.*

behavior across workloads. `TinySTORMER-LG` exhibits lower fidelity (79.5%), reflecting the impact of gradient accumulation and accelerated computation on proxy accuracy. This occurs because gradient accumulation accelerates computation and reduces synchronization frequency, which changes the overlap between computation and I/O. Such behavior is difficult to faithfully emulate in the proxy because it depends on asynchronous GPU execution and vendor-specific synchronization mechanisms. Despite this variability, the proxy preserves relative performance trends across configurations, which is sufficient to guide optimization.

Subset Representativeness. To reduce optimization cost, we evaluate configurations using reduced dataset subsets. As shown in **Table 4**, Full-to-Subset fidelity remains consistently high across all workloads and exceeds 97%. This indicates that appropriately chosen subsets preserve full-scale performance behavior while substantially reducing execution cost during optimization. As mentioned in Section 4.4 and illustrated in **Figure 4**, smaller subsets fail to saturate storage and memory subsystems. Subset configurations are chosen per workload to balance cost reduction and fidelity. As shown in **Table 5**, subset ratios vary across applications, yet consistently reduce per-configuration evaluation time by 1.6 to 8.5×.

Proxy-to-App Transfer Accuracy. After optimizing on the proxy and subset configuration, selected parameters are evaluated on the original application. **Table 4** shows that most workloads exceed 90% transfer accuracy, except `SFNO-LG` with lower accuracy of 54.6%. This discrepancy arises from differences in execution balance between the proxy and the full application. After optimization, `SFNO-LG` exhibits improved overlap between computation and I/O, altering resource contention and execution timing in ways not captured by proxy execution. Because the proxy incurs minimal data-fetch overhead, it underestimates the absolute performance gains achieved at full scale. A detailed execution-time breakdown for `SFNO-LG` is provided in **Appendix ??**. Despite this divergence in absolute magnitude, Section 5.7 shows that optimized configuration improves application throughput relative to original configuration, indicating that the proxy preserves the relative performance ordering needed to guide effective optimization.

Aggregate Accuracy. Across all workloads, App-to-Proxy accuracy averages 93.7%, Full-to-Subset accuracy averages 98%, and Proxy-to-App accuracy averages 86.7%. While fidelity varies by workload, proxy-based approximation consistently preserves relative optimization trends across the configuration space.

Takeaway 3. *GPU-free proxies preserve original workload behavior and relative improvement while making optimization time practical.*

Table 5: GLANCED-IO achieves up to 8.5× reduction in optimization cost via *Subset Generation*.

Workload	Subset	Full (5 ep)	Subset (5 ep)	Speedup
STORMER-LG	1/2	~84 min	~35 min	2.4×
SFNO-LG	1/4	~120 min	~27 min	4.4×
TinySTORMER-LG	1/2	~86 min	~54 min	1.6×
STORMER-SM	1/4	~57 min	~13 min	4.3×
SFNO-SM	1/8	~60 min	~7 min	8.5×

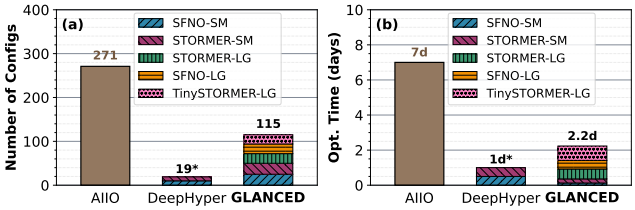


Figure 5: Optimization cost comparison.GLANCED-IO reduces (a) number of configurations to evaluate by up to 2.3× and (b) optimization time by up to 3.2×, enabling practical deployment-time tuning without prior data collection. *DeepHyper cannot be run at larger scale due to storage limitations.

5.5 Optimization Cost

Having established that proxy-based approximation preserves optimization fidelity while reducing per-configuration cost, we now evaluate the efficiency and scalability of GLANCED-IO's optimization for large configuration spaces. To understand this, we run *Guided Optimizer* across all workloads defined in Section 5.1.3. We exclude jobs queuing and dataset generation time to ensure fairness.

As detailed in Section 5.1.4, we compare GLANCED-IO with AIIO and DeepHyper. Additionaly, we used data collection time for building of AIIO models as their optimization cost. **Figure 5** compares the number of configurations evaluated and end-to-end optimization time between GLANCED-IO, AIIO, and DeepHyper, where y-axis in (a) shows the number of configurations and (b) shows optimization time in days. GLANCED-IO reduces the number of evaluated configurations by 2.3× compared to AIIO using Greedy-OFAT exploration that scales linearly with the number of tunable parameters rather than exhaustively searching the configuration space. This reduction directly translates into optimization time savings: AIIO requires 7 days, including upfront profiling and prediction-guided search, whereas GLANCED-IO completes in 2.2 days by combining Greedy-OFAT with proxy-based approximation and subset evaluation (**§5.4**).

Takeaway 4. *GLANCED-IO achieves 3.2× faster optimization than ML-based approach by eliminating upfront data collections.*

At small scale, each workload has 1,568 possible configurations. DeepHyper, constrained to 12-hour budgets per workload, explores only 10 and 9 configurations for `SFNO-SM` and `STORMER-SM` respectively (~0.6% each), totaling 19 evaluations in 24 hours due to full-pipeline overhead. In contrast, GLANCED-IO's data-centric subsetting evaluates 25 configurations per workload (50 total) within 7.2 hours, a 3.3× speedup over DeepHyper while exploring 2.6× more configurations. At large scale, DeepHyper is not practical

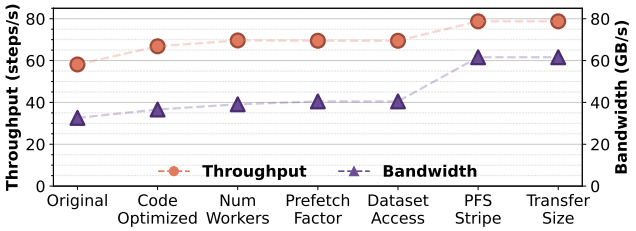


Figure 6: STORMER-LG optimization.GLANCED-IO simultaneously improves both application and system layer by 1.3×.

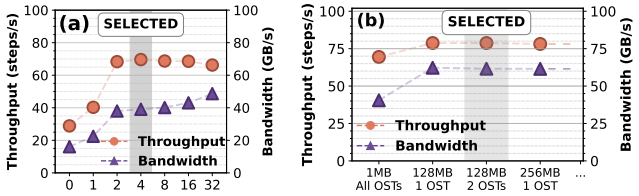


Figure 7: Stage-level optimizations for STORMER-LG. ***(a)*** *GLANCED-IO choose number of workers with the highest throughput (~69 steps/s), avoiding degradation of application performance in larger number of workers.* ***(b)*** *Selected stripe configuration improves I/O bandwidth by 1.3×.*

in our setting because the search space includes not only runtime knobs, but also *dataset configurations*. Thus, evaluating a candidate may require building a new dataset variant rather than toggling a parameter, and precomputing all variants would exceed our 100TB budget by reaching petabyte-scale storage.

On-the-fly generation does not fundamentally solve the problem as it shifts the overhead into the search loop. Trials must wait for data generation, limiting the number of configurations that can be explored within the time budget. GLANCED-IO avoids this issue through OFAT which materializes data transformations incrementally. Furthermore the subset-based optimization reduces storage in direct proportion to subset size; for example, a quarter-sized subset requires only a quarter of the storage.

Takeaway 5. *Storage is a hidden optimization cost: upfront data generation can exceed budgets before search begins.*

5.6 End-to-End Optimization: STORMER-LG

To illustrate how proxy-guided joint optimization translates into end-to-end application performance gains, we present STORMER-LG as a representative large-scale, I/O-bound case study. This workload uses a large number of GPUs and a multi-terabyte dataset, exhibiting substantial I/O pressure and a large configuration space.

Optimization Trajectory. Figure 6 shows optimization trajectory for STORMER-LG. Across stages, both application throughput and I/O bandwidth improve monotonically. The optimized configuration achieves a 1.3× improvement in application throughput and I/O bandwidth relative to original application, demonstrating consistent gains at both application and system levels.

Application-level Parameter Tuning. Figure 7a shows the effect of tuning application-level parameters. Increasing the number of data loader workers initially improves application throughput, but performance peaks at a small worker count and degrades beyond that point due to CPU contention. In contrast, I/O bandwidth continues to increase with additional workers. Configurations that maximize bandwidth alone therefore lead to reduced end-to-end throughput, whereas configurations near the throughput peak achieve better joint performance. Prefetch factor tuning has limited impact for this workload, and the original dataset layout is retained.

System-level Parameter Tuning. Figure 7b shows the impact of system-level tuning. Moving away from the default Lustre striping configuration yields substantial improvements in both application throughput and I/O bandwidth. Among the evaluated settings, a configuration using two OSTs with a 128 MB stripe size provides the best joint performance. Further tuning of transfer size does not improve performance and is therefore not selected.

Optimization Cost. For STORMER-LG, optimization is performed using a 1/2 dataset subset, reducing per-configuration evaluation time from approximately 84 minutes to 35 minutes (**Table 5**). In total, GLANCED-IO evaluates only 22 configurations across all optimization stages, and the complete end-to-end optimization finishes in ~13 hours, as shown in **Figure 5b**. In contrast, exhaustive exploration of the full configuration space would require evaluating orders of magnitude more configurations, making it infeasible at this scale, whereas GLANCED-IO remains modest relative to large-scale training runs and enables efficient optimization in deployment.

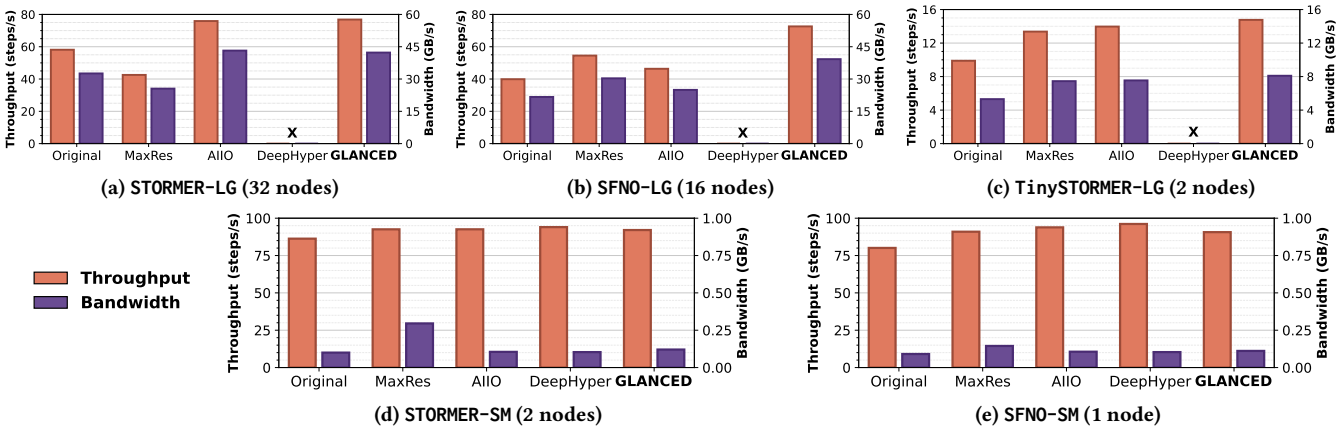
End-to-end Outcome. The final optimized configuration improves application throughput while maintaining high system utilization. As shown in **Figure 8**, the optimized STORMER-LG configuration achieves the highest application throughput among the evaluated approaches, reflecting the benefit of multi-objective optimization. A detailed baseline comparison is discussed in Section 5.7.

Takeaway 6. *The* STORMER-LG *case study confirms that proxy- and subset-based joint optimization yields substantial end-to-end performance gains at scale while keeping optimization overhead practical.*

5.7 Baseline Comparison

To evaluate GLANCED-IO's end-to-end effectiveness relative to existing approaches, we compare it against three representative baselines: MaxResource, AIIO, and DeepHyper detailed in Section 5.1.4. After getting configurations from these baselines, we run each workload with selected configurations and compare it to GLANCED-IO. To ensure fairness, each system used the same code-optimized version of the application as its starting point. **Figure 8** reports application throughput and I/O bandwidth for the configuration selected by each approach. In the figures, left y-axis is the throughput shown using orange bar and right y-axis is the I/O bandwidth shown using purple bar. To make comparison easier, we also include original pipeline performance as the leftmost bars on each workload.

Large-scale, I/O-bound workloads (STORMER-LG, SFNO-LG). GLANCED-IO consistently achieves the highest application throughput while maintaining high I/O bandwidth. AIIO attains comparable bandwidth and competitive throughput but with 3.2× slower optimization time (**§5.5**). However, for SFNO-LG, AIIO fails to identify effective configuration, resulting in 1.57× lower application throughput and I/O bandwidth compared to GLANCED-IO. This occurs because AIIO selects configurations that maximize I/O bandwidth without accounting for cross-layer interactions, leading to choices



(a) STORMER-LG (32 nodes) (b) SFNO-LG (16 nodes) (c) TinySTORMER-LG (2 nodes)

(d) STORMER-SM (2 nodes) (e) SFNO-SM (1 node)

Figure 8: Baseline comparison. *GLANCED-IO matches or outperforms AIIO and DeepHyper (small-scale only) performance with 3.2× and 3.3× faster optimization respectively, outperforms MaxRes at scale, and achieves up to 1.8× gains over original pipelines.* ***X*** *= infeasible due to storage constraints.*

that do not translate into end-to-end performance gains. MaxResource underperforms in large-scale workloads despite allocating the maximum available resources, emphasizing that maximizing resource utilization does not guarantee better performance. DeepHyper is infeasible at large scale due to storage limitations (**§5.5**).

Small-scale workloads (STORMER-SM, SFNO-SM). For smaller workloads, performance differences are reduced because application throughput is already near peak and I/O pressure is limited. DeepHyper achieves similar results to GLANCED-IO in this regime but with 3.3× slower optimization time, while AIIO and MaxResource also remain competitive. Importantly, GLANCED-IO does not degrade performance in these regimes, indicating robust behavior when optimization headroom is constrained. On average, GLANCED-IO achieves up to 1.8× improvement in both application throughput and I/O bandwidth compared to original pipelines.

Takeaway 7. *Joint optimization improves end-to-end performance for I/O-bound workloads, while avoiding performance regressions when I/O is not the dominant bottleneck.*

5.8 Cross-Workload Optimization Behavior

Having established optimization fidelity and cost effectiveness, we examine how GLANCED-IO performs across workloads with different I/O regimes. Rather than repeating full optimization trajectories, we summarize recurring patterns that emerge across workloads.

Large-scale, I/O-bound Workloads (STORMER-LG, SFNO-LG). For workloads operating under sustained I/O pressure, joint optimization consistently yields substantial performance improvements. In both STORMER-LG and SFNO-LG, increasing I/O parallelism, such as the number of data loader workers, improves system utilization but degrades application throughput. The unified metric reliably identifies configurations that balance these effects, avoiding high-bandwidth configurations that yield low end-to-end throughput (**Figure 8a,b**). On both workloads, filesystem striping provides the largest system-level performance gains, even when the most effective application-level knobs vary across workloads (**Figure 7b**). Specifically for SFNO-LG, GLANCED-IO improves the original pipeline by 1.8× in both application throughput and I/O bandwidth, outperforming all other baselines.

Compute-accelerated Workload (TinySTORMER-LG). TinySTORMER-LG exhibits different behavior due to accelerated computation from gradient accumulation, which increases pressure on the data-loading pipeline. In this regime, the optimizer selects more aggressive application-level parallelism, such as a higher number of data loader workers, to keep pace with computation, while dataset reorganization provides little benefit. Despite altered execution characteristics, the optimization process converges to configurations that improve both application throughput and I/O bandwidth.

Non-I/O-bound Workload (STORMER-SM, SFNO-SM). For small workloads, application throughput is already near peak and largely insensitive to I/O tuning. Many configuration changes increase measured I/O bandwidth without improving throughput, and some actively degrade performance. For example, increasing transfer size substantially improves bandwidth while reducing application throughput (**Figure 1b**). The unified metric correctly avoids such configurations by accounting for application-level impact, and optimization converges quickly without regressions.

6 Related Work

GLANCED-IO differs from prior works in three key aspects: objective metrics, configuration tuning, and proxy construction.

6.1 Optimization Metrics

Existing works optimize HPC and DL pipelines by adopting a single metric as the optimization objective. Tools such as DYAD [32], HVAC [40], noPFS [37], and Accelerating Data Loading [57] focused on optimizing data loading, such as enabling locality-aware data loading, to optimize application throughput. Meanwhile, tf-Darshan [31], DeepIO [62], Continuous Characterization via Darshan [30] and DFTracer [33] focused on speeding up data loading from system layer using I/O metrics as primary objectives. All of the above approaches are complementary to and remain applicable within GLANCED-IO; however, GLANCED-IO jointly accounts for both performance metrics to ensure that optimizing one does not degrade the other. More recent work, such as DLIO [34], highlighted the benefits of jointly considering both application throughput and I/O bandwidth. Similarly, MLPerf Storage [4], a widely adopted

community standard, requires submissions to maintain at least 90% of application throughput, even though final rankings are based solely on I/O bandwidth. Together, these efforts indicate a growing recognition that I/O performance should be optimized without degrading application performance, further motivating GLANCED-IO to use joint metrics during optimization.

6.2 Configuration Tunings

Prior works address the explosion of configuration space during I/O optimization. For example, H5Tuner [22, 25] uses evolutionary algorithms to navigate large configuration spaces and tune multi-layer parameters at runtime. In contrast, HEPnOS Autotuning [36] adopts Bayesian optimization via DeepHyper [21] to optimize a distributed storage service tailored for physics applications. With the rise of machine learning, approaches such as AIIO [35] aim to eliminate exhaustive search altogether, while TunIO [50] leverages reinforcement learning to identify high-impact configurations. Lastly, DLIO [34] uses controlled parameter sweeping during optimization as a proof that controlled optimization on high impact configuration will still be beneficial while keeping the cost minimal. GLANCED-IO avoids limitations above by using OFAT-guided greedy optimization, which keeps the configuration space linear while prioritizing high-impact parameters.

6.3 Proxy Applications

Many works reduce evaluation cost by constructing lightweight proxies of the original applications during optimization. In HPC, several I/O kernels, such as VPIC-IO [10, 28], VORPAL-IO [48], and GCRM-IO [51], are manually extracted from application codebases. This extraction process can be automated using Automatic Kernels Generation [23]. In the DL domain, tools such as DLIO [34] and GPEmu [54] model end-to-end pipelines and enable low-cost GPU-free evaluation. To further reduce evaluation cost, works in the ML community, including AUTOMATA [42] and Two-Step Hyperparameter Optimization [60], select representative data subsets to accelerate model tuning. Mantevo [44] develops workload-specific proxy applications (miniapps) that preserve the behavior of various HPC applications for general benchmarking studies. In contrast, GLANCED-IO uses GPU-free proxy and lightweight subset construction to support efficient I/O optimization across DL workloads.

7 Conclusion

With the growing scale of data-intensive DL workloads on HPC systems, I/O has become a dominant bottleneck as its behavior varies widely across applications and execution regimes. GLANCED-IO provides a cross-layer framework that considers both application performance and system utilization to optimize I/O in distributed training pipelines. Our results show that optimizing only application performance can lead to underutilized system resources, while optimizing only I/O performance can degrade end-to-end application throughput, leaving up to 2.4× performance unrealized. We further demonstrate that the cost of autotuning and ML-based approaches often outweighs their performance benefits in practice, as they incur high optimization overhead or require upfront data collection. In contrast, GLANCED-IO's OFAT-guided greedy exploration achieves performance comparable to state-of-the-art techniques while remaining inexpensive, and its proxy-based optimization using representative data subsets enables GPU-free evaluation while preserving 93% average performance fidelity. Across five global weather forecasting workloads, GLANCED-IO improves performance by up to 1.57× over baselines, evaluates 2.3× fewer configurations than AIIO, and completes optimization 3.3× faster than DeepHyper. We further demonstrate that GLANCED-IO generalizes across diverse workloads and execution regimes, identifying balanced I/O parallelism in I/O-bound cases while avoiding performance degradation when I/O is not the bottleneck. As future work, we will leverage GLANCED-IO's efficient evaluation to explore iterative refinement and stochastic exploration for further performance gains.

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